Aditya Mittha

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Summary

Final-year Electronics and Telecommunication engineering student (9.27 CGPA) with proven hands-on expertise in Embedded Software Engineering, Firmware Architecture, and Low-Level Driver Development. Experienced in writing modular Embedded C/C++ for ARM Cortex-M and ESP32 microcontrollers, designing FreeRTOS multi-threaded firmware, and implementing communication protocol stacks (UART, SPI, I2C, CAN, MQTT). Skilled in interrupt service routines (ISR), circular ring buffers, hardware abstraction layers (HAL), and debugging with logic analyzers and JTAG/SWD.

Skills

Languages & Core: Embedded C (C99/C11), C++, Python (Automation/Testing), Bash Scripting, MISRA C Concepts

RTOS & Firmware: FreeRTOS (Tasks, Queues, Semaphores, Mutexes), Bare-Metal Systems, State Machines (FSM), HAL/BSP

Protocols & Interfaces: UART, SPI, I2C, CAN Bus (Arbitration/Frames), MQTT (QoS), GPIO, ADC, PWM, Timers

Hardware & Debugging: ESP32 (ESP-IDF), ARM Cortex-M (STM32), Raspberry Pi, Logic Analyzer, GDB, JTAG/SWD

DevOps & Tools: Git, GitHub Actions, Linux (Ubuntu/WSL, Shell, Make/CMake), Docker, AWS IoT Core

Soft Skills: Analytical Debugging, Systems Architecture, Clean Code Practices, Cross-Functional Collaboration

Education

Bachelor's Degree in Electronics and Telecommunication 2027
Walchand Institute Of Technology *CGPA: 9.27*

Higher Schooling 2023
D.H.B Soni College, Solapur *Percentage: 83%*

Work Experience

Embedded Software Developer Intern May 2025 – June 2025
Codec Technologies India *Remote*

- Authored modular, driver-level Embedded C firmware for UART and I^2C communication peripherals, integrating circular ring buffers that eliminated packet loss.
- Programmed Hardware Abstraction Layer (HAL) routines and Interrupt Service Routines (ISRs) on microcontrollers for synchronized, low-latency sensor data acquisition.
- Debugged bus timing bottlenecks and arbitration conflicts using logic analyzers and serial debuggers, resolving 10+ firmware defects prior to deployment.
- Maintained structured code reviews, version control, and development documentation across active Git branches in a Linux environment.

Technologies: Embedded C, UART, I^2C, ISRs, HAL, Logic Analyzer, Git, Linux

Projects

AQUANOVA – ESP32 Real-Time Firmware & Telemetry Engine 2026

- Architected multi-threaded FreeRTOS firmware on ESP32 managing non-blocking ADC/GPIO polling, queue-based IPC, and asynchronous MQTT telemetry dispatch.
- Designed resilient peripheral drivers with hardware watchdog timer monitoring and automatic reconnect state machines for robust network failover.
- Engineered circular ring buffers handling high-frequency sensor streams with zero buffer overflow during real-time pressure spikes.

Role: Embedded Software & Firmware Lead (3-person team)
Tech Stack: Embedded C, ESP32, FreeRTOS, MQTT, I^2C, ADC, Watchdog Timer, Python, Git

LabPulse – Embedded Telemetry Agent & Interfacing System 2026

- Developed a modular telemetry agent service in Python to monitor peripheral hardware status, session states, and serial data streams with sub-1% CPU overhead.
- Implemented secure REST API communication with SHA-256 validation to transmit device diagnostics to AWS serverless backends (Lambda, DynamoDB).
- Containerized background services with Docker and set up automated CI/CD pipelines in GitHub Actions.

Role: Backend Lead & System Software Architecture
Tech Stack: Python, Embedded Linux, REST APIs, AWS Lambda, DynamoDB, Docker, GitHub Actions

Rewards & Achievements

- 2x Winner, National Electronic Quiz Competition (Quizotronics)
- Runner-up, State-level chess championship
- Participated in 4^+ Hackathons and Embedded Hardware Competitions

Languages

English | Hindi | Marathi | Kannada